

Supplementary Information

Crop-ranking reversal under downside risk and operational constraints

Xinyu Lu and Peimin Chen

Contents

1	Definitions and complete model notation	1
2	Full finite-scenario linear programme	2
2.1	KKT equations	3
3	Proofs	3
3.1	Proof of the restricted reversal theorem	3
3.2	Proof of complete optimality	3
3.3	Proof of named-family ordering	3
3.4	Proofs for information and flexibility	4
4	Calibration and constraint table	4
5	Policy comparison	5
6	Phase-diagram summary	5
7	Robustness and uncertainty	5
8	Information–flexibility details	6
9	External panel and identification boundary	6
10	Data availability	6
11	Code availability	6
12	Limitations	6

1 Definitions and complete model notation

Pairwise reversal is $s_i > s_j$, $x_i < x_j$. Complete rank reversal means that the top-ranked crop receives less than every lower-ranked crop. Strong reversal uses exclusion: $s_i > s_j$, $x_i = 0 < x_j$. These definitions separate a strict allocation-order inversion from complete exclusion of the higher-ranked crop.

Symbol	Definition
$i = 1, \dots, n$	crop index
ω	joint price–yield–cost state
s_i	external pre-decision agronomic or predictive score

Symbol	Definition
x_i	planted area or normalized land share
$\tilde{P}_i, \tilde{Y}_i, \tilde{C}_i$	stochastic price, yield and cost
$\tilde{\pi}_i$	stochastic per-acre margin $\tilde{P}_i \tilde{Y}_i - \tilde{C}_i$
μ_i	expected per-acre margin
$L(x)$	portfolio loss $-\tilde{\pi}^\top x$
A	available land
B, c	working-capital limit and crop cash coefficients
R, r	rotation rows and limits
K, k	contractual-minimum rows and requirements
G, h	shared-capacity rows and limits
ℓ, u	crop-specific lower and upper area bounds
α	CVaR confidence level
κ	loss-CVaR ceiling
ρ	normalized risk-tolerance path index
$\mathcal{X}$	operational feasible set
$\mathcal{S}_\kappa$	complete optimal set under ceiling κ
σ	deterministic selector for one reported optimum
θ	parameter along a named joint-law family
ϕ	operational-flexibility or buffering parameter

2 Full finite-scenario linear programme

For scenario probabilities $w_m > 0$ summing to one, the implemented primal is

$$\max_{x,v,q} \quad \mu^\top x \tag{1}$$

$$\text{s.t.} \quad \mathbf{1}^\top x \leq A, \tag{2}$$

$$c^\top x \leq B, \tag{3}$$

$$Rx \leq r, \quad -Kx \leq -k, \quad Gx \leq h, \tag{4}$$

$$\ell \leq x \leq u, \tag{5}$$

$$v + \frac{1}{1-\alpha} \sum_{m=1}^M w_m q_m \leq \kappa, \tag{6}$$

$$-\pi_m^\top x - v - q_m \leq 0, \quad m = 1, \dots, M, \tag{7}$$

$$q_m \geq 0, \quad v \in \mathbb{R}. \tag{8}$$

There is no hidden penalty term and no post-solution adjustment of the optimized policy. Heuristic benchmark policies are projected onto the feasible set so that all policies are compared inside the same operational set.

2.1 KKT equations

Let η be the CVaR-ceiling multiplier, t_m the multiplier for scenario excess row m , and z_m the multiplier on $q_m \geq 0$. Together with the operational multipliers defined in the main text, stationarity is

$$0 = -\mu + \gamma \mathbf{1} + \beta c + R^\top \rho - K^\top \chi + G^\top \lambda - a + b - \sum_m t_m \pi_m, \quad (9)$$

$$0 = \eta - \sum_m t_m, \quad (10)$$

$$0 = \frac{\eta w_m}{1 - \alpha} - t_m - z_m, \quad m = 1, \dots, M. \quad (11)$$

All multipliers are nonnegative except the dual expression associated with the free variable v . Complementarity requires each multiplier times its primal slack to be zero. The scenario equations allow fractional mass at the VaR threshold and are valid for discrete scenario distributions.

3 Proofs

3.1 Proof of the restricted reversal theorem

Let $z = x_1$ and $x_2 = A - z$. Expected profit is

$$\mu_1 z + \mu_2 (A - z) = \mu_2 A + (\mu_1 - \mu_2) z.$$

Because $\mu_1 > \mu_2$, this function is strictly increasing in z . Loss-CVaR is convex and continuous in the portfolio weights. Its sublevel set intersected with the compact operational interval I is therefore a compact interval $\mathcal{K}_{\theta, \kappa}$. A strictly increasing linear objective has the unique maximizer $\sup \mathcal{K}_{\theta, \kappa}$.

The pair reverses when $z < A - z$, equivalently $z < A/2$. Hence reversal holds if and only if the right endpoint of the entire risk-feasible interval lies below $A/2$, which is equivalent to no feasible point on $[A/2, \bar{z}]$. If risk is nondecreasing on that upper half, the minimum upper-half risk occurs at $A/2$, so the upper half is infeasible if and only if $r_\theta(A/2) > \kappa$. Existence of a feasible lower-half point ensures that this inequality indicates reversal rather than infeasibility.

If $\theta \mapsto r_\theta(A/2)$ is continuous, strictly increasing and crosses κ , the intermediate-value theorem gives a unique θ^* . If strictness is removed, an interval of thresholds can result. If monotonicity is removed, the reversal set can be disconnected. If crossing is removed, a threshold need not exist.

3.2 Proof of complete optimality

The finite-scenario problem is a bounded linear programme whenever $\mathcal{X}$ is nonempty and bounded. Linear-programming strong duality implies that primal and dual feasibility plus equality of the objective values are necessary and sufficient. Expanding equality of values across all inequality rows yields stationarity and complementarity. The equations above therefore form a complete certificate.

For the population convex formulation, write the Lagrangian using loss-CVaR as a convex function. Under Slater's condition or another convex constraint qualification, KKT conditions are necessary; convexity makes them sufficient. Pairwise subtraction of acreage stationarity cancels γ but leaves budget, rotation, contract, shared-resource and bound terms. No algebraic operation on this local equality recovers an acreage ordering without the global feasible geometry.

3.3 Proof of named-family ordering

For integrable random losses, CVaR is a law-invariant coherent risk measure and is monotone under convex order. Thus $L_{\theta_1}(x) \preceq_{\text{cx}} L_{\theta_2}(x)$ implies the stated CVaR inequality at each x . Pointwise inequality gives containment of the two risk sublevel sets. Maximizing the same expected-profit objective over nested sets

gives the optimal-value order. Coordinate monotonicity does not follow because the supporting face can change when the feasible set contracts.

3.4 Proofs for information and flexibility

In a posterior-separable model, or when the shared ex-ante risk restriction is nonbinding, any no-information action defines a constant signal-contingent policy. If the constant policy is feasible, optimization after observing a signal cannot lower value. When one common action is posterior-optimal for every signal, the constant policy also attains the informed optimum, proving zero value. If unique posterior optima differ on events with positive probability and at least one difference is strict, expected improvement is positive. This posterior-optimum strictness step is not applied when contingent actions share a binding ex-ante CVaR budget.

Increasing differences and nested action sets yield monotone posterior actions under standard monotone comparative statics. Strictness on a positive- probability event makes the information gain larger at the higher acreage- flexibility level. This is the conditional complementarity result.

For buffering, at $\phi = 1$ all state margins equal $\bar{\pi}$; posterior optimization problems are identical and the common-action condition gives zero information value. If information value is positive at $\phi = 0$ and continuous along the contraction path, it must decline somewhere. Those intervals are substitutive. Total optimized value may still rise with ϕ .

For the complete shared-CVaR numerical programme, interaction is classified without extending that theorem:

$$\Delta_{q,\phi}V = [V(q_2, \phi_2) - V(q_1, \phi_2)] - [V(q_2, \phi_1) - V(q_1, \phi_1)].$$

The sign is computed on each adjacent grid rectangle.

4 Calibration and constraint table

Table 2: Kansas calibration and primary allocation.

Crop	score	mean margin real 2024 US\$ per acre	margin s.d.	full-model share
Corn	0.748	199.8	94.9	0.330
Soybean	0.756	229.0	104.0	0.422
Winter wheat	0.894	138.3	49.6	0.248

Table 3: Operational set. Coefficients use normalized land and capacity units.

Object	Corn	Soybean	Winter wheat
Lower bound	0.05	0.10	0.05
Upper bound	0.55	0.65	0.60
Rotation cap	0.50	—	—
Contract minimum	—	0.10	—
Planting-labour coefficient	1.00	0.72	0.58
Harvest-equipment coefficient	1.05	0.82	0.62

Total land is 1.0, operating budget is US\$330, planting-labour capacity is 0.82 and harvest-equipment capacity is 0.88. The full model uses $\alpha = 0.95$, $\rho = 0.5$, 512 optimization scenarios and seed 202607270034.

Table 4: Primary policy outcomes. A larger loss-CVaR is worse.

Policy	Corn	Soy	Wheat	Profit	loss-CVaR	risk feasible
Suitability proportional	0.312	0.315	0.373	187.4	-53.1	yes
Winner take all, projected	0.300	0.100	0.600	166.6	-57.3	yes
Equal share	0.333	0.333	0.333	190.4	-51.4	yes
Expected profit	0.280	0.650	0.070	216.5	-34.5	no
Mean-variance	0.280	0.650	0.070	216.5	-34.5	no
Full loss-CVaR	0.330	0.422	0.248	198.5	-47.0	yes
Minimum-CVaR endpoint	0.220	0.180	0.600	169.0	-59.5	yes

5 Policy comparison

The primary policy table retains a low-penalty mean-variance diagnostic for comparability. The diversification comparison is separate. Its x^0 is the matched-Gaussian expected-profit allocation; all 301 points over $\gamma \in [0, 0.03]$ are solved; and its selected x^{MV} is the first point to attain the fixed 15% Gaussian variance-reduction target. At that point the reduction is 15.3%, the allocation differs from x^T , and Student- t evaluation loss-CVaR is \$2.364 higher. The weak and strong criteria hold over γ intervals 0.0068-0.0088 and 0.0068-0.0088. Because x^T binds the common ceiling, the tail-inferiority and ceiling-violation gaps are numerically dependent.

One-factor sensitivity contains 18 cases covering scenario count, seed, Kendall’s τ , Student- t copula degrees of freedom, CVaR confidence level, risk-ceiling location, evaluation-law marginal specification and the 10%, 15% and 20% selection targets. 17 cases satisfy the selected-policy weak and strong criteria. The retained null is the 20% target, which selects $\gamma = 0.0089$, just beyond the focal strong-failure interval; its positive tail gap is smaller than the declared risk tolerance ε_r .

6 Phase-diagram summary

The mechanism-isolation outputs retain every evaluated cell. In the margin layer, top-ranked winter wheat has the lowest public-data mean margin and the expected-profit endpoint is already completely inverted; CVaR raises its share. In the risk layer, Soybean has the higher score and mean margin, while a mean-preserving 10% downside event of \$154.7 per acre produces the loose-to-tight allocation crossing reported in the main text. Across the probability-by-magnitude map, 24 of 49 cells cross, 24 remain feasible without crossing and 1 are infeasible. In the operational layer, the observed constraint-addition sequence and its null crossing result precede the complete soybean rotation-cap grid; the first crossing is at 0.35. These controlled paths do not estimate the prevalence of either mechanism.

Across all 165 cells, selected pairwise reversal occurs in 143, selected complete rank reversal in 96 and strong exclusion reversal in 0. Complete family-by- τ table, all three ranked-pair face extrema and possible, universal and selected classifications are retained for every cell.

7 Robustness and uncertainty

The one-at-a-time robustness file reports selected pairwise, selected complete and selected strong classifications separately for confidence level, constraint regime, dependence family and parameter, marginal law, sample window, scenario count, random seed and solver. With positive crop lower bounds, none of these cells can satisfy the exclusion definition.

The 64 historical bootstrap replications produce mean shares 0.277 for corn, 0.527 for soybean and 0.196 for winter wheat. Their 95% percentile intervals are [0.125, 0.368], [0.258, 0.650] and [0.070, 0.404]. Reversal probability is treated separately: selected pairwise reversal occurs in 62 of 64 resamples. Its

exact Clopper–Pearson binomial interval is [0.892,0.996]; this is not a percentile bootstrap interval. The first-reversal index mean is 0.066 with interval [0.000,0.598]. The broad boundary interval is consistent with an eight-year calibration and is not hidden by the large number of simulated scenarios.

8 Information–flexibility details

The state split contains four low and four high relative-yield-advantage years. Signal accuracy takes six values and flexibility takes six values on each path, producing 72 cells. Adjacent-grid classifications are 32 positive cross-differences, 17 negative cross-differences, 18 zero-information and 5 zero/boundary. Every signal-contingent problem is feasible and the minimum computed value of information differs from zero only at floating-point scale.

At perfect signal accuracy, post-signal acreage reallocation raises information value monotonically from zero at no adjustable acreage to \$4.655 at full adjustment. Under shock buffering, the corresponding information value declines toward zero as the state margins are contracted to a common vector. This comparison is about interaction; buffering can still increase the no-information optimized value.

9 External panel and identification boundary

Table 5: Top-rank disagreement in 248 state-years.

Ranking definition	estimate	cluster 95% low	cluster 95% high
Relative yield	0.887	0.802	0.954
Standardized revenue	0.613	0.447	0.767
Operating margin	0.625	0.552	0.693
Total-cost margin	0.782	0.701	0.857

The leakage-free relative-yield specification uses 651 crop-transition rows. Its prior-leader effect is -0.0042 with interval [-0.0123,0.0056]. The estimand is descriptive because the panel does not observe farm objectives, local prices or costs, credit, rotation history, contracts, risk preferences or received forecasts. State acreage is never substituted for an optimizer output.

10 Data availability

The analysis uses public USDA NASS, USDA ERS and US Bureau of Labor Statistics files joined by crop and year. The public source reports, analysis-ready tables and figure source data needed to reproduce the reported results will be deposited in a versioned public archive upon publication. No confidential farm-level data are used.

11 Code availability

Analysis code sufficient to regenerate the numerical results and figures will be deposited with the source data upon publication.

12 Limitations

1. The Kansas calibration has eight effective annual observations. It cannot estimate extreme-tail dependence reliably.

2. State yield combined with national price and cost is not a farm-level profit series.
3. Copula parameters and shared-capacity limits are controlled stress values, not causal estimates.
4. The model is single-period and linear. It omits fixed charges, endogenous prices, spatial spillovers and explicit multi-year rotation state.
5. Historical bootstrap replications quantify sensitivity to the short calibration record but do not create independent empirical observations.
6. The information state is a transparent annual proxy, not a deployed forecast product.
7. The external panel is aggregate and descriptive; it cannot identify the full model's active mechanism or welfare.